

1. Find your Public IP Address

- Search "What is my IP" on Google.

Aim

To find the public IP address of the device connected to the internet.

Steps to Find Public IP Address

1. Open any web browser such as Google Chrome, Microsoft Edge, or Firefox.
2. Go to Google Search.
3. Type “**What is my IP**” in the search bar.
4. Press Enter.
5. Google will display your public IP address at the top of the page.

Observation

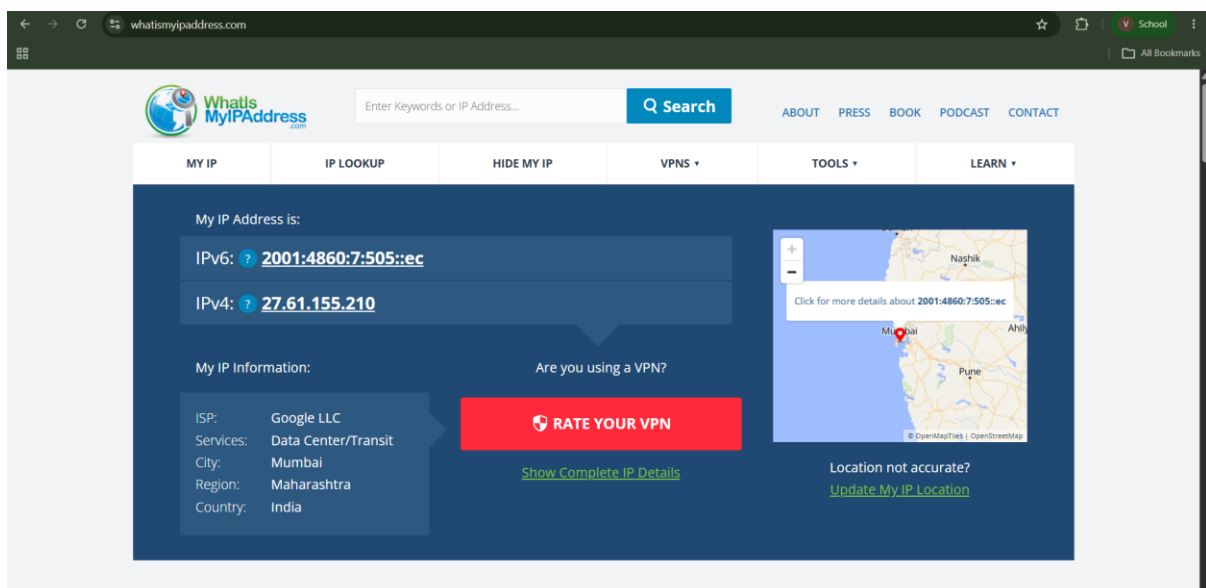
The displayed IP address is known as the **Public IP Address**. This address is assigned by the Internet Service Provider (ISP) and is used to identify your network on the internet.

Example

Example Public IP Address: 49.xxx.xxx.xxx

(Note: Your actual IP address may be different.)

Screenshot



2. Find the IP Address of the following websites:

- **google.com**
- **facebook.com**
- **amazon.com**

Aim

To identify the IP addresses of popular websites.

Method Used

The IP address of a website can be found using:

- Command Prompt (CMD)
- Ping command
- DNS lookup tools

Steps Using Command Prompt

1. Open Command Prompt.
2. Type the command:
ping google.com
3. Press Enter.
4. The IP address will be displayed in brackets.

Repeat the same process for other websites.

Explanation

- Every website on the internet has an IP address.
- DNS (Domain Name System) converts website names into IP addresses.
- IP addresses help computers locate and communicate with servers on the internet.

For google.com:

```
C:\Users\vaish>ping google.com

Pinging google.com [2404:6800:4009:812::200e] with 32 bytes of data:
Reply from 2404:6800:4009:812::200e: time=36ms
Reply from 2404:6800:4009:812::200e: time=36ms
Reply from 2404:6800:4009:812::200e: time=30ms
Reply from 2404:6800:4009:812::200e: time=31ms

Ping statistics for 2404:6800:4009:812::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 30ms, Maximum = 36ms, Average = 33ms

C:\Users\vaish>
```

The screenshot shows the result of the command: ping google.com

The system resolved the domain name **google.com** into the IP address:
2404:6800:4009:812::200e

This is an **IPv6 address** used by Google servers.

What is Ping?

The ping command is a network utility used to test connectivity between a computer and another device or website over the internet.

It helps to:

- Check whether a website/server is reachable
- Measure response time
- Detect packet loss
- Test network connectivity

Details from the Screenshot

- **Packets Sent:** 4
- **Packets Received:** 4
- **Packet Loss:** 0%
- **Average Response Time:** 33 ms

This indicates that the connection to Google was successful and stable.

Understanding the IP Address

The address:

2404:6800:4009:812::200e

is an IPv6 address because:

- It contains hexadecimal characters

- Uses colons (:) instead of dots (.)
- Provides a much larger address space compared to IPv4

For facebook.com:

```
C:\Users\vaish>ping facebook.com

Pinging facebook.com [2a03:2880:f379:1:face:b00c:0:25de] with 32 bytes of data:
Reply from 2a03:2880:f379:1:face:b00c:0:25de: time=37ms
Reply from 2a03:2880:f379:1:face:b00c:0:25de: time=49ms
Reply from 2a03:2880:f379:1:face:b00c:0:25de: time=37ms
Reply from 2a03:2880:f379:1:face:b00c:0:25de: time=41ms

Ping statistics for 2a03:2880:f379:1:face:b00c:0:25de:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 37ms, Maximum = 49ms, Average = 41ms
```

The screenshot shows the result of the command: ping facebook.com

The domain name **facebook.com** was converted into the IP address:

2a03:2880:f379:1:face:b00c:0:25de

This is an IPv6 address used by Facebook servers.

Analysis of Ping Result

The ping command tested the connectivity between the computer and Facebook's server.

The results show:

- **Packets Sent:** 4
- **Packets Received:** 4
- **Packet Loss:** 0%
- **Minimum Time:** 37 ms
- **Maximum Time:** 49 ms
- **Average Time:** 41 ms

This means:

- The internet connection is working properly.
- Facebook's server is reachable.
- No packets were lost during communication.

What the Response Time Means

The response time (measured in milliseconds) indicates how quickly the server replied to the request.

Lower response time means:

- Faster connection
- Better network performance

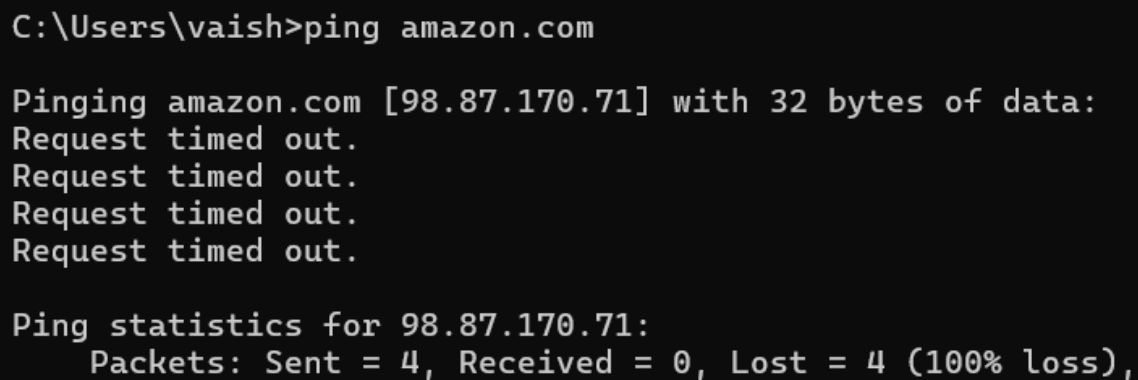
IPv6 Address Explanation

The address: 2a03:2880:f379:1:face:b00c:0:25de

is an IPv6 address because:

- It contains hexadecimal numbers and letters.
- It uses colon (:) separators.
- It supports a larger number of devices compared to IPv4.

For amazon.com:



```
C:\Users\vaish>ping amazon.com

Pinging amazon.com [98.87.170.71] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 98.87.170.71:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Explanation of Amazon Ping Output

The screenshot shows the result of the command:

ping amazon.com

The domain name **amazon.com** was resolved into the IP address:

98.87.170.71

This is an IPv4 address used by Amazon servers.

Analysis of Ping Result

The ping request was sent to Amazon's server, but all requests timed out.

The results show:

- Packets Sent: 4
- Packets Received: 0
- Packet Loss: 100%

This means the system was able to find Amazon's IP address, but the server did not respond to the ping requests.

Reason for Request Timed Out

Many large websites such as Amazon block or ignore ping requests for security reasons. Even though the website is working normally in a browser, it may not reply to ICMP ping packets.

IPv4 Address Explanation

The address:

98.87.170.71

is an IPv4 address because:

- It contains four decimal numbers separated by dots.
- IPv4 addresses are the most commonly used IP addresses on the internet.

3. Explain the difference between:

- HTTP and HTTPS
- Public IP and Private IP

Difference Between HTTP and HTTPS

HTTP	HTTPS
Stands for HyperText Transfer Protocol	Stands for HyperText Transfer Protocol Secure
Used for communication between browser and website	Used for secure communication between browser and website
Data is transferred in plain text	Data is transferred in encrypted form
Less secure	More secure
Vulnerable to hacking and data theft	Protects data from hackers and cyber attacks
Does not use SSL/TLS certificate	Uses SSL/TLS certificate
URL starts with http://	URL starts with https://
Uses Port 80	Uses Port 443
No data encryption	Encrypts usernames, passwords, and banking information
Slower security level	Higher security level
Mainly used for informational websites	Used for banking, shopping, login, and secure websites
No padlock icon shown in browser	Padlock icon shown in browser
Cannot verify website authenticity	Verifies website authenticity
Easier for attackers to intercept communication	Difficult for attackers to intercept communication
Does not provide confidentiality	Provides confidentiality and integrity

Difference Between Public IP and Private IP

Public IP Address	Private IP Address
Used for communication over the internet	Used within a local network
Assigned by Internet Service Provider (ISP)	Assigned by router or network administrator
Globally unique	Can be reused in different networks
Accessible from anywhere on the internet	Not directly accessible from the internet
Used by websites and internet servers	Used by devices inside homes, schools, and offices
Helps identify a network on the internet	Helps identify devices inside a local network
Requires ISP management	Managed locally
More exposed to cyber threats	More secure because it is internal
Used for external communication	Used for internal communication
Example: 27.61.155.210	Example: 192.168.1.1
Limited number of available addresses	Can be reused multiple times
Connected directly to the internet	Connected through routers/NAT
Used by modems and routers	Used by computers, mobiles, printers, etc.
Publicly visible	Hidden from public internet
Needed for online hosting and communication	Needed for local device communication

4. Open any 5 websites and check:

- **Do they use HTTPS?**
- **What company owns the website?**

Website 1 – Google

Website:google.com

Uses HTTPS?

Yes, Google uses HTTPS.

Proof:

The URL starts with: <https://www.google.com>

A lock icon is also visible in the browser address bar, which indicates a secure HTTPS connection.

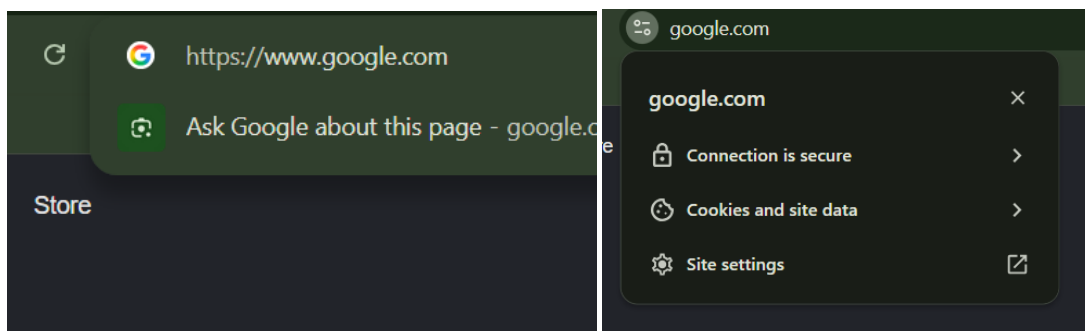
Company Owner:Google LLC

About the Company:

Google LLC is an American multinational technology company. It provides services such as:

- Search Engine
- Gmail
- Google Maps
- YouTube
- Cloud Services

Screenshot:



Website 2 – Facebook

Website:facebook.com

Uses HTTPS?

Yes, Facebook uses HTTPS.

Proof:

The URL starts with: <https://www.facebook.com>

The browser displays a lock icon, showing that the website connection is secure.

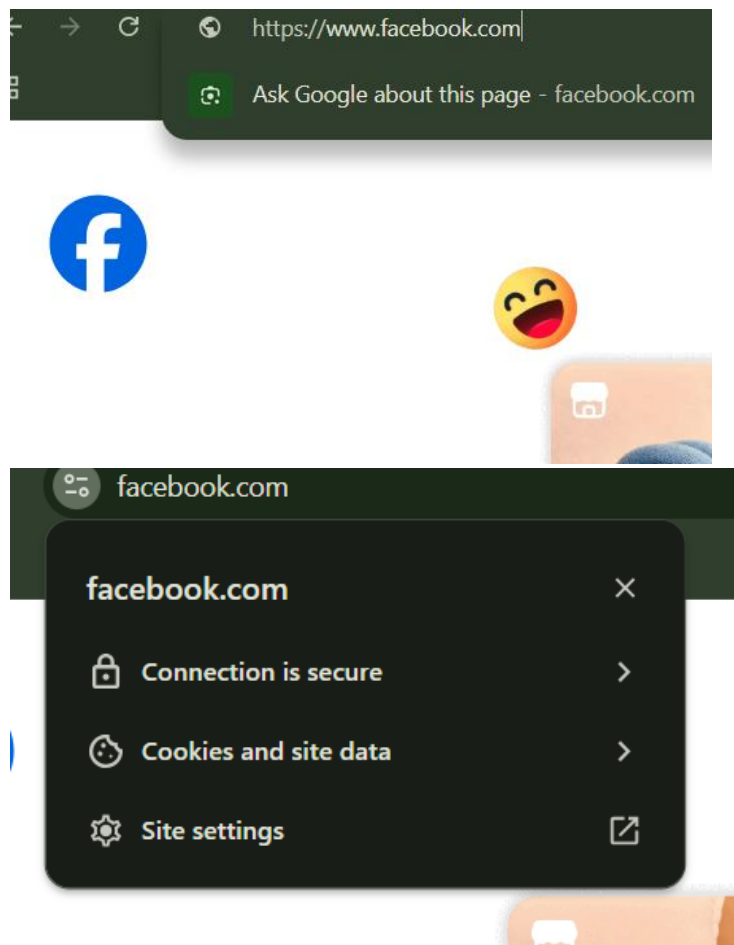
Company Owner:Meta Platforms Inc.

About the Company:

Meta Platforms Inc. is a social media and technology company that owns:

- Facebook
- Instagram
- WhatsApp
- Messenger

Screenshot:



Website 3 – Amazon

Website:amazon.com

Uses HTTPS?

Yes, Amazon uses HTTPS.

Proof:

The URL starts with: <https://www.amazon.com>

The secure lock icon confirms encrypted communication.

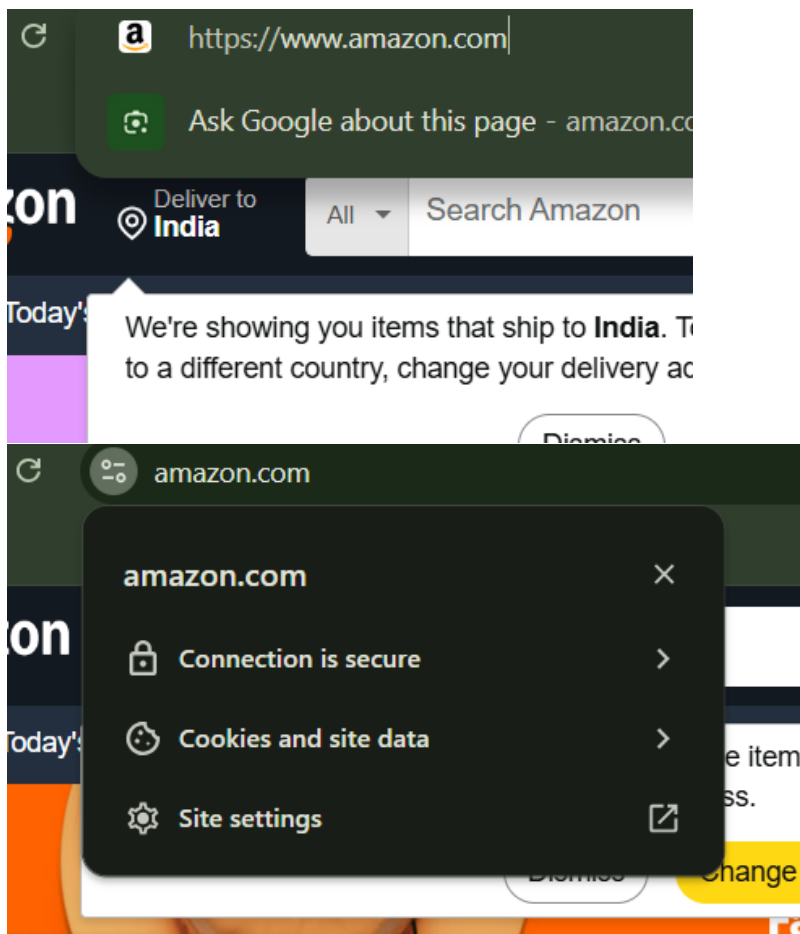
Company Owner:Amazon.com Inc.

About the Company:

Amazon is a multinational company known for:

- Online Shopping
- Cloud Computing
- Digital Streaming
- Artificial Intelligence Services

Screenshot:



Website 4 – YouTube

Website: youtube.com

Uses HTTPS?

Yes, YouTube uses HTTPS.

Proof:

The URL starts with: <https://www.youtube.com>

The lock symbol in the browser indicates a secure website.

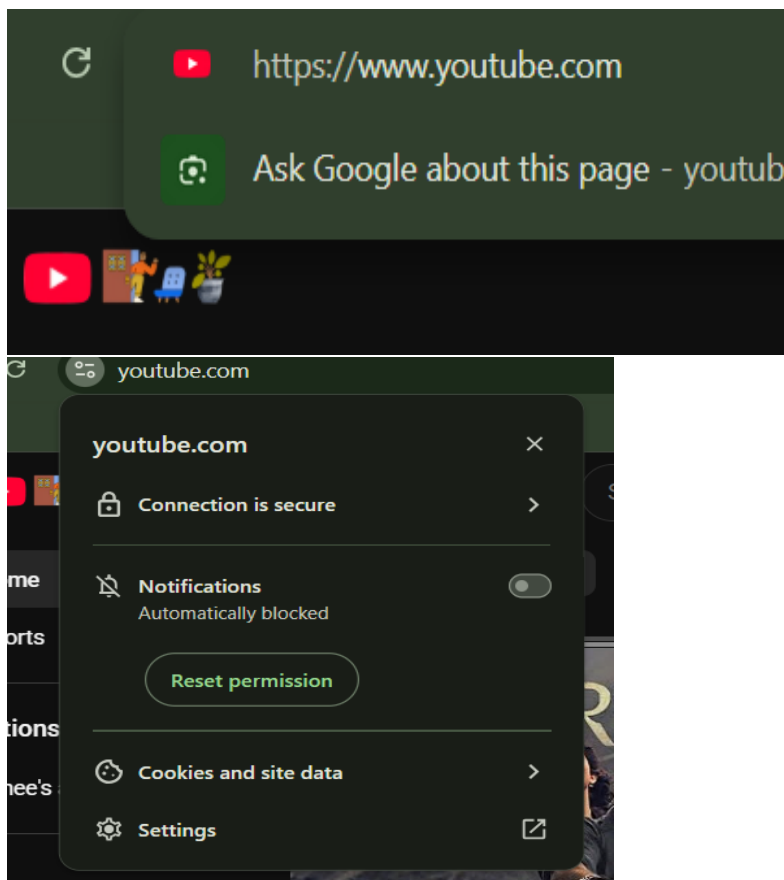
Company Owner: Google LLC

About the Company:

YouTube is a video-sharing platform where users can:

- Upload videos
- Watch videos
- Stream live content
- Share media online

Screenshot:



Website 5 – Wikipedia

Website: wikipedia.org

Uses HTTPS?

Yes, Wikipedia uses HTTPS.

Proof:

The URL starts with: <https://www.wikipedia.org>

The website displays a secure HTTPS connection.

Company Owner: Wikimedia Foundation

About the Company:

Wikimedia Foundation is a non-profit organization that manages:

- Wikipedia
- Wikimedia Commons
- Wiktionary
- Other educational projects

Screenshot:

